Flutter Lab Programs

Short, complete programs — Questions 1–17

Required packages: Add `shared_preferences` for programs 4–11, 14, 15 and 17; add `path_provider` for programs 12, 13 and 16. Run each Flutter program separately as `lib/main.dart`.

1. Store student details in a Map

```
void main() {
  Map<String, dynamic> student = {
    'name': 'Harsh', 'roll': 101, 'course': 'BCA'
  };
  student.forEach((key, value) => print('$key: $value'));
}
```

Output: name: Harsh | roll: 101 | course: BCA

2. Convert Student object to Map and back

```
class Student {
  String name;
  int roll;
  Student(this.name, this.roll);

  Map<String, dynamic> toMap() =>
      {'name': name, 'roll': roll};

  factory Student.fromMap(Map<String, dynamic> m) =>
      Student(m['name'], m['roll']);
}

void main() {
  var s = Student('Harsh', 101);
  var map = s.toMap();
  var s2 = Student.fromMap(map);
  print(map);
  print('${s2.name} ${s2.roll}');
}
```

Output: {name: Harsh, roll: 101} | Harsh 101

3. Convert Map to JSON and JSON to Map

```
import 'dart:convert';

void main() {
  Map<String, dynamic> data = {'name': 'Harsh', 'age': 20};
  String json = jsonEncode(data);
  Map result = jsonDecode(json);
  print(json);
  print(result);
}
```

Output: {"name":"Harsh","age":20} | {name: Harsh, age: 20}

4. Save and display user name using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Home()));

class Home extends StatefulWidget {
  @override
  State<Home> createState() => _HomeState();
}
class _HomeState extends State<Home> {
  final c = TextEditingController();
  String name = '';
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => name = p.getString('name') ?? '');
  }
  Future save() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('name', c.text);
    load();
  }
  @override
```

```
  Widget build(BuildContext context) => Scaffold(
    appBar: AppBar(title: Text('Save Name')),
    body: Column(children: [
      TextField(controller: c),
      ElevatedButton(onPressed: save, child: Text('Save')),
      Text('Welcome $name')
    ]),
  );
}
```

Output: Saved name is displayed after reopening the app.

5. Store and retrieve counter using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Counter()));
class Counter extends StatefulWidget {
  @override
  State<Counter> createState() => _CounterState();
}
class _CounterState extends State<Counter> {
  int n = 0;
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => n = p.getInt('count') ?? 0);
  }
  Future add() async {
    final p = await SharedPreferences.getInstance();
    await p.setInt('count', ++n);
    setState(() {});
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Counter')),
    body: Center(child: Column(mainAxisSize: MainAxisSize.min,
      children: [Text('$n'), ElevatedButton(onPressed: add,
      child: Text('Add'))])),
  );
}
```

Output: Counter increases and its value remains saved after reopening.

6. Save Dark Mode setting using Switch

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(App());
class App extends StatefulWidget {
  @override
  State<App> createState() => _AppState();
}
class _AppState extends State<App> {
  bool dark = false;
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => dark = p.getBool('dark') ?? false);
  }
  Future change(bool v) async {
    final p = await SharedPreferences.getInstance();
    await p.setBool('dark', v);
    setState(() => dark = v);
  }
  @override
  Widget build(BuildContext c) => MaterialApp(
    theme: dark ? ThemeData.dark() : ThemeData.light(),
    home: Scaffold(appBar: AppBar(title: Text('Dark Mode')),
      body: Switch(value: dark, onChanged: change)),
  );
}
```

Output: Switch changes the theme; the choice is saved.

7. Remember Me checkbox using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Login()));
class Login extends StatefulWidget {
```

```
  @override
  State<Login> createState() => _LoginState();
}
class _LoginState extends State<Login> {
  bool remember = false;
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => remember = p.getBool('remember') ?? false);
  }
  Future save(bool v) async {
    final p = await SharedPreferences.getInstance();
    await p.setBool('remember', v);
    setState(() => remember = v);
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Login')),
    body: CheckboxListTile(title: Text('Remember Me'),
      value: remember, onChanged: (v) => save(v!)),
  );
}
```

Output: Remember Me selection remains saved after reopening.

8. Save name, email and age using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Profile()));
class Profile extends StatefulWidget {
  @override
  State<Profile> createState() => _ProfileState();
}
class _ProfileState extends State<Profile> {
  final name = TextEditingController(), email = TextEditingController();
  final age = TextEditingController();
  String result = '';
  Future save() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('name', name.text);
    await p.setString('email', email.text);
    await p.setInt('age', int.tryParse(age.text) ?? 0);
    setState(() => result =
      '${name.text}, ${email.text}, ${age.text}');
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('User Details')),
    body: ListView(padding: EdgeInsets.all(16), children: [
      TextField(controller: name, decoration: InputDecoration(labelText: 'Name')),
      TextField(controller: email, decoration: InputDecoration(labelText: 'Email')),
      TextField(controller: age, keyboardType: TextInputType.number,
        decoration: InputDecoration(labelText: 'Age')),
      ElevatedButton(onPressed: save, child: Text('Save')),
      Text(result)
    ]),
  );
}
```

Output: Entered name, email and age are saved and displayed.

9. Store and display a list of strings

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Subjects()));
class Subjects extends StatefulWidget {
  @override
  State<Subjects> createState() => _SubjectsState();
}
class _SubjectsState extends State<Subjects> {
  List<String> items = [];
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => items = p.getStringList('subjects') ?? []);
  }
  Future save() async {
    final p = await SharedPreferences.getInstance();
    items = ['Dart', 'Flutter', 'DBMS'];
    await p.setStringList('subjects', items);
```

```
    load();
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Subjects')),
    body: Column(children: [
      ElevatedButton(onPressed: save, child: Text('Save Subjects')),
      ...items.map((e) => ListTile(title: Text(e)))
    ]),
  );
}
```

Output: Dart, Flutter and DBMS appear in the list.

10. Clear SharedPreferences data

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: ClearData()));
class ClearData extends StatelessWidget {
  Future clear() async {
    final p = await SharedPreferences.getInstance();
    await p.clear();
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Clear Data')),
    body: Center(child: ElevatedButton(
      onPressed: clear, child: Text('Clear All Data'))),
  );
}
```

Output: Pressing Clear All Data removes saved preferences.

11. Show welcome screen only on first launch

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Welcome()));
class Welcome extends StatefulWidget {
  @override
  State<Welcome> createState() => _WelcomeState();
}
class _WelcomeState extends State<Welcome> {
  String msg = '';
  @override
  void initState() { super.initState(); check(); }
  Future check() async {
    final p = await SharedPreferences.getInstance();
    bool first = p.getBool('first') ?? true;
    setState(() => msg = first ? 'Welcome!' : 'Welcome Back!');
    await p.setBool('first', false);
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    body: Center(child: Text(msg)),
  );
}
```

Output: First launch shows Welcome!; later launches show Welcome Back!

12. Write text into a file using path_provider

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

void main() => runApp(MaterialApp(home: FileWrite()));
class FileWrite extends StatelessWidget {
  Future write() async {
    final d = await getApplicationDocumentsDirectory();
    final f = File('${d.path}/note.txt');
    await f.writeAsString('Hello Flutter!');
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Write File')),
    body: Center(child: ElevatedButton(
      onPressed: write, child: Text('Write Text'))),
  );
}
```

Output: Text is written to note.txt in the documents directory.

13. Read text from a stored file

```
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

void main() => runApp(MaterialApp(home: FileRead()));
class FileRead extends StatefulWidget {
  @override
  State<FileRead> createState() => _FileReadState();
}
class _FileReadState extends State<FileRead> {
  String data = '';
  Future read() async {
    final d = await getApplicationDocumentsDirectory();
    final f = File('${d.path}/note.txt');
    data = await f.exists() ? await f.readAsString() : 'File not found';
    setState(() {});
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Read File')),
    body: Column(children: [
      ElevatedButton(onPressed: read, child: Text('Read')),
      Text(data)
    ]),
  );
}
```

Output: Displays the text stored in note.txt.

14. Save Student object as JSON in SharedPreferences

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: SaveStudent()));
class SaveStudent extends StatefulWidget {
  @override
  State<SaveStudent> createState() => _SaveStudentState();
}
class _SaveStudentState extends State<SaveStudent> {
  String result = '';
  Future save() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('student',
      jsonEncode({'name': 'Harsh', 'roll': 101}));
    result = jsonDecode(p.getString('student')!) .toString();
    setState(() {});
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Student JSON')),
    body: Column(children: [
      ElevatedButton(onPressed: save, child: Text('Save & Display')),
      Text(result)
    ]),
  );
}
```

Output: {name: Harsh, roll: 101}

15. To-Do app using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: Todo()));
class Todo extends StatefulWidget {
  @override
  State<Todo> createState() => _TodoState();
}
class _TodoState extends State<Todo> {
  final c = TextEditingController();
  List<String> tasks = [];
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => tasks = p.getStringList('tasks') ?? []);
  }
  Future add() async {
    tasks.add(c.text);
    final p = await SharedPreferences.getInstance();
    await p.setStringList('tasks', tasks);
```

```
    c.clear(); load();
  }
  @override
  Widget build(BuildContext cxt) => Scaffold(
    appBar: AppBar(title: Text('To-Do')),
    body: Column(children: [
      TextField(controller: c),
      ElevatedButton(onPressed: add, child: Text('Add')),
      Expanded(child: ListView(
        children: tasks.map((t) => ListTile(title: Text(t))).toList()))
    ]),
  );
}
```

Output: Added tasks are shown and restored when the app is reopened.

16. Notes app using a JSON file

```
import 'dart:convert';
import 'dart:io';
import 'package:flutter/material.dart';
import 'package:path_provider/path_provider.dart';

void main() => runApp(MaterialApp(home: Notes()));
class Notes extends StatefulWidget {
  @override
  State<Notes> createState() => _NotesState();
}
class _NotesState extends State<Notes> {
  final c = TextEditingController();
  List<String> notes = [];
  Future<File> get file async {
    final d = await getApplicationDocumentsDirectory();
    return File('${d.path}/notes.json');
  }
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final f = await file;
    if (await f.exists()) notes = List<String>.from(jsonDecode(await f.readAsString()));
    setState(() {});
  }
  Future save() async {
    notes.add(c.text);
    final f = await file;
    await f.writeAsString(jsonEncode(notes));
    c.clear(); load();
  }
  @override
  Widget build(BuildContext cxt) => Scaffold(
    appBar: AppBar(title: Text('Notes')),
    body: Column(children: [
      TextField(controller: c),
      ElevatedButton(onPressed: save, child: Text('Save Note')),
      Expanded(child: ListView(
        children: notes.map((n) => ListTile(title: Text(n))).toList()))
    ]),
  );
}
```

Output: Notes are saved in notes.json and loaded when the app starts.

17. Student Profile app using SharedPreferences

```
import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MaterialApp(home: StudentProfile()));
class StudentProfile extends StatefulWidget {
  @override
  State<StudentProfile> createState() => _StudentProfileState();
}
class _StudentProfileState extends State<StudentProfile> {
  final n = TextEditingController(), r = TextEditingController();
  final co = TextEditingController(), s = TextEditingController();
  String profile = '';
  @override
  void initState() { super.initState(); load(); }
  Future load() async {
    final p = await SharedPreferences.getInstance();
    setState(() => profile = p.getString('profile') ?? '');
  }
  Future save() async {
    final p = await SharedPreferences.getInstance();
    await p.setString('profile',
      '${n.text}, ${r.text}, ${co.text}, ${s.text}');
```

```
    load();
  }
  Future clear() async {
    final p = await SharedPreferences.getInstance();
    await p.remove('profile');
    n.clear(); r.clear(); co.clear(); s.clear(); load();
  }
  @override
  Widget build(BuildContext c) => Scaffold(
    appBar: AppBar(title: Text('Student Profile')),
    body: ListView(padding: EdgeInsets.all(12), children: [
      for (final x in [n, r, co, s])
        TextField(controller: x),
      ElevatedButton(onPressed: save, child: Text('Save / Edit')),
      ElevatedButton(onPressed: clear, child: Text('Clear')),
      Text(profile)
    ]),
  );
}
```

Output: Saved student profile is displayed; it can be edited or cleared.